Metric Spaces

The idea of metric and metric space are abstractions of the concept of distance in Euclidean space.

<u>Defn</u>: A metric on a set X is a function $d: X \times X \to \mathbb{R}$ that satisfies the following conditions:

(a) $d(x,y) \geq 0$ for all $x, y \in X$.

(b) $d(x,y) = 0$ iff $x = y$

(c) $d(x,y) = d(y,x)$ for all $x, y \in X$.

(d) $d(x,z) \leq d(x,y) + d(y,z)$ for all $x, y, z \in X$.

If d is a metric on X, the ordered pair (X,d) is called the metric space.

<u>Remark</u>: Property (d) of a metric is an abstraction of the fact that the length of one side of a triangle is less than or equal to the sum of the lengths of the other two sides.

<u>Examples</u>:

1. $d: \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ defined by $d(x,y) = |x-y|$. It is called the usual metric on $\mathbb{R}$.

$d: \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ $\quad d((x_1,y_1),(x_2,y_2)) = \left[(x_1-x_2)^2 + (y_1-y_2)^2\right]^{1/2}$.

Let X be a set.

2. $d: X \times X \to \mathbb{R}$ by $d(x,x) = 0$. $\forall x \in X$

$d(x,y) = 1$, if $x, y \in X$ & $x \neq y$.

It is called the discrete metric

3. $d : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $d((x,y)) = \left(\sum (x_i - y_i)^2\right)^{1/2}$

It is called the usual metric for $\mathbb{R}^n$.

If we take $X = \mathbb{R}$ in example 2, then we have two metrics on $\mathbb{R}$, the usual metric & the discrete metric.

4. Let $n \in \mathbb{N}$ and define $d : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ by

$$d(x,y) = \sum_{i=1}^{N} |x_i - y_i|$$

Then d is a metric on $\mathbb{R}^n$, and it is called the Taxicab metric.

5. Let $n \in \mathbb{N}$ and define a function $\rho : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ by

$$\rho(x,y) = \max \{ |x_1 - y_1|, |x_2 - y_2|, \dots, |x_n - y_n| \}$$

where $x = (x_1, x_2, \dots, x_n)$
$y = (y_1, y_2, \dots, y_n)$.

Then ρ is a metric on $\mathbb{R}^n$, & it is called the square metric.

pf. $\rho(x,y) \geq 0 \ \forall\, x, y \in \mathbb{R}^n$, $\rho(x,y) = 0$ iff $x = y$.

& $\rho(x,y) = \rho(y,x)$

For each $i = 1, 2, \dots, n$, $|x_i - z_i| \leq |x_i - y_i| + |y_i - z_i|$

Therefore, by definition of ρ, $|x_i - z_i| \leq \rho(x,y) + \rho(y,z)$ for each $i = 1, 2, \dots, n$. Hence $\rho(x,z) \leq \rho(x,y) + \rho(y,z)$.

Remark: For $\mathbb{R}^1$, the Euclidean metric, the taxicab metric and the square metric are precisely the same.

Recall

Theorem: Let $x=(x_1,\dots,x_n)$ and $y=(y_1,\dots,y_n)\in\mathbb{R}^n$. Then

$$|x\cdot y| \le \|x\|\,\|y\|.$$

Example: 6. Let $X=\ell^2 := \{x=(x_1,\dots,x_n,\dots) \in : \sum |x_i|^2<\infty\}$.

$$d(x,y) = \left(\sum_{i=1}^{\infty}(x_i-y_i)^2\right)^{1/2}$$ Then d is a well-defined metric on ℓ^2, and it is called ℓ^2-metric.

7. Let $a,b\in\mathbb{R}$ with $a<b$, and let X denote the set of all continuous functions on $[a,b]$, $X=C[a,b]$. Define a function

$d: X\times X\to\mathbb{R}$ by

$$d(f,g)=\int_a^b |f(x)-g(x)|\,dx.$$ Then (X,d) is a metric space.

pf.

$$d(f,g)\ge 0, \quad d(f,g)=d(g,f).$$

$$|f(x)-h(x)| \le |f(x)-g(x)|+|g(x)-h(x)|$$

$$\Rightarrow \quad d(f,h) \le d(f,g)+d(g,h)$$

It remains to show: $d(f,g)=0 \Rightarrow f\equiv g$.

$$\int_a^b |f(x)-g(x)|\,dx = 0 \Rightarrow f\equiv g \text{ i.e. } f(x)=g(x)\ \forall x\in[a,b].$$

Continuous function:

Let (X, d) and (Y, ρ) be metric spaces. A function $f: X \to Y$ is continuous at a point "a" in X provided for each $\varepsilon > 0$, there is a positive number $\delta > 0$ s.t if $x \in X$ and $d(a, x) < \delta$

$$\Rightarrow \quad \rho\big(f(a), f(x)\big) < \varepsilon.$$

A function $f: X \to Y$ is continuous provided it is continuous at each point of X.

Example: Let (X, d) & (Y, ρ) be metric spaces, and let $y \in Y$. $i: X \to X$ defined by $i(x) = x$ and $c: X \to Y$ defined by $c(x) = c_0 \; \forall x \in X$ are continuous.

Indeed, Let $a \in X$ & let $\varepsilon > 0$. For any $\varepsilon > 0$, choose $\delta = \varepsilon$..

Then if $x \in X$ & $d(a, x) < \delta$, $d(i(a), i(x)) = d(a, x) < \varepsilon$.

$\Rightarrow$ i is continuous.

Let (X, d) be a metric space, let $\varepsilon > 0$ be a positive real number, & let $x \in X$ The set:

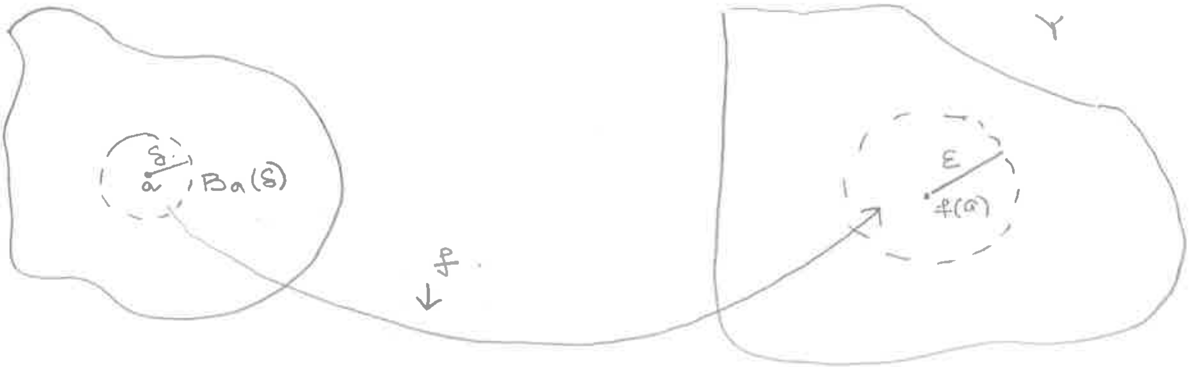
$$B_d(x, \varepsilon) = \{ y \in X : d(x, y) < \varepsilon \}$$

is called the ε-ball with center at x.

The definition of continuity in metric space can be rephrased in terms of open balls:

Theorem: Let (X, d) & (Y, ρ) be metric spaces. A function $f: X \to Y$ is continuous at a point "a" in X iff for each open ball, $B_\rho(f(a), \varepsilon)$, there is an open ball $B_d(a, \delta)$, s.t $f(B_d(a, \delta)) \subseteq B_\rho(f(a); \varepsilon)$.

pf.



Defn: A subset U of a metric space (X,d) is open if for each $x \in U$, there is a open ball $B_d(x,\varepsilon)$ s.t $B_d(x,\varepsilon) \subseteq U$.

Theorem: The open subsets of a metric space (X,d) have the following properties:

(a) X & ϕ are open sets

(b) The union of any collection of open sets is open.

(c) The intersection of any finite collection of open sets is open.

pf. (a) Let $x \in X$, and let $\varepsilon > 0$. Then $B_d(x,\varepsilon) \subseteq X$ & hence X is open. The ϕ is open because there is no $x \in \phi$.

(b) Let $\{U_\alpha : \alpha \in \Lambda\}$ be a collection of open sets and let $x \in \bigcup_{\alpha \in \Lambda} U_\alpha \Rightarrow \exists\ \beta \in \Lambda$ s.t $x \in U_\beta$.

Since U_β is open, there exists an open Ball $B_d(x,\varepsilon)$ s.t $B_d(x,\varepsilon) \subseteq U_\beta$. Since $U_\beta \subseteq \bigcup U_\alpha$

$\Rightarrow B_d(x,\varepsilon) \subseteq \bigcup_{\alpha \in \Lambda} U_\alpha \Rightarrow \bigcup U_\alpha$ is open.

(c) ex R.

Remark: Notice that arbitrary intersection of open sets need not be open. Let $U_n = \left(-\frac{1}{n}, \frac{1}{n}\right)$ in $\in \mathbb{N}$. Then each U_n is an open subset of $\mathbb{R}$, but $\bigcap_{n\in\mathbb{N}} U_n = \{0\}$. & $\{0\}$ is not open in $\mathbb{R}$.

Lemma: An open non-empty subset of $\mathbb{R}$ (with usual metric) is any set that can be expressed as the union of open intervals.

pf: U is an open subset of $\mathbb{R}$, & let $x \in U$. Then there exists $\varepsilon > 0$ s.t $B_d(x,\varepsilon) \subseteq U$. $B_d(x,\varepsilon) = (x-\varepsilon, x+\varepsilon)$.

Therefore U can be expressed as open interval.

Example: Let X be a non-empty set, and let d be the discrete metric on X. Then for each $x \in X$ & each ε with $0 < \varepsilon \leq 1$, $B_d(x,\varepsilon) = \{x\}$ for each x,

$\Rightarrow$ $\{x\}$ is an open subset of (X, d).

Remark: $\{0\}$ is not open in $\mathbb{R}$ with usual metric, but it is open in $\mathbb{R}$ with respect to another metric.

Another definition of Continuity:

Let $f: (X,d) \to (Y,\rho)$ be a function, and let $a \in X$. Then f is continuous at a iff for each open subset V of Y s.t $f(a) \in V$, there is an open subset U of X s.t $a \in U$ & $f(U) \subseteq V$.

§ Topological Spaces:

The open sets in a metric space have the properties listed in the previous thm, and these properties are used to define a topology on a set.

Defn: A topology on a set X is a collection $\mathcal{T}$ of subsets of X having the following properties:

(a) $\phi \in \mathcal{T}$ & $X \in \mathcal{T}$.

(b) If $U \in \mathcal{T}$ and $V \in \mathcal{T}$, then $U \cap V \in \mathcal{T}$.

(c) If $U_\alpha \in \mathcal{T}$ for each α in an index set Λ, then $\cup \{U_\alpha : \alpha \in \Lambda\} \in \mathcal{T}$.

The members of $\mathcal{T}$ are called open sets.

Remark Let $\mathcal{T}$ be a topology on a set X and let Λ be a finite set. If $U_\alpha \in \mathcal{T}$ for each $\alpha \in \Lambda$, then

$$\bigcap_{\alpha \in \Lambda} U_\alpha \in \mathcal{T}$$

Defn: A topological space is an ordered pair $(X, \mathcal{T})$, where X is a set & $\mathcal{T}$ is a topology on X.

Examples

1. Let (X,d) be a metric. The open subsets of (X,d) form a topology on X. This topology is called the topology on X induced by d. Thus every metric space is a topological space.

Defn: A topological space $(X, \mathcal{T})$ is metrizable if there is a metric d on X such that the topo. induced by d ~~on~~ is $\mathcal{T}$.

2. Let X be a set with atleast two members, and let $\mathcal{T}^{i}$ be the trivial topology on X. Then $(X, \mathcal{T}^{i})$ is not metrizable.

pf: Let d be a metric on X & let $\mathcal{U}$ be the topology generated by d. We claim that $(X, \mathcal{T}^{i})$ is not metrizable by showing that $\mathcal{U} \neq \mathcal{T}^{i}$. Let a & b denote distinct members of X then $d(a,b) = r$ (there is a positive number r).

Therefore $a \in B_d(a;r)$, but $b \notin B_d(a;r)$.

$\Rightarrow B_d(a;r) \in \mathcal{U}$ but $B_d(a;r) \neq X$, and $B_d(a;r) \neq \phi$.

Thus $\mathcal{U} \neq \mathcal{T}^{i}$.

3. Let X be a non-empty set, and let $\mathcal{T}^{D}$ denotes the collection of all subsets of X $(\mathcal{T}^{D} = \mathcal{P}(X))$. Then $\mathcal{T}^{D}$ is a topology on X, and it is called discrete topology on X.

Remark: The discrete metric on a non-empty set X induced the discrete top. on X.

4. Let $\mathcal{T}^{u} = \{ U \in \mathcal{P}(\mathbb{R}) : \text{for each } x \in U, \text{there is a positive number } \delta_x \text{ s.t. } (x-\delta_x, x+\delta_x) \subseteq U \}$.

Then $\mathcal{T}^{u}$ is the topology on $\mathbb{R}$ induced by the usual metric.

5. Let $\mathcal{T} = \{ U \in \mathcal{P}(\mathbb{R}^2) :$ for each $(a,b) \in U$, there is a positive number δ s.t $\{ (x-a)^2 + (y-b)^2 < \delta^2 \} \subseteq U \}$.

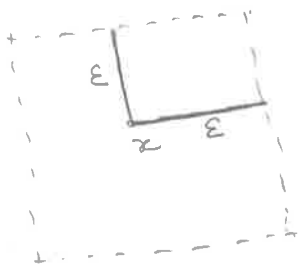
$\mathcal{T}$ is the topo. on $\mathbb{R}^2$ induced by usual metric on $\mathbb{R}^2$.

Two metrics d & ρ on a set X induce same topology on X provided that a subset of X is open in (X,d) iff it is open in (X,ρ).

6. The topology generated by the square metric ρ on $\mathbb{R}^2$ is the usual topology.

$$\rho(x,y) = \max\{ |x_1 - y_1| , |x_2 - y_2| \} <$$

$$B_\rho(x,\varepsilon) = \{ y \in \mathbb{R}^2 : \rho(x,y) < \varepsilon \}$$



Let d be the usual metric on $\mathbb{R}^2$. Let U be an open subset of $(\mathbb{R}^2, \rho)$, and let $x \in U$, then there exists a positive number ε s.t $B_\rho(x,\varepsilon) \subseteq U$.

$$\rho(a,b) \leq d(a,b) \Rightarrow B_d(x,\varepsilon) \subseteq B_\rho(x,\varepsilon).$$

$\Rightarrow B_d(x,\varepsilon) \subseteq U$ & hence U is an open in $(\mathbb{R}^2, d)$.

let U be an open in $(\mathbb{R}^2, d)$, and let $x \in U$

$\Rightarrow \exists\, \varepsilon > 0$ s.t $B_d(x,\varepsilon) \subseteq U$.

Since $d(a,b) \leq \sqrt{2} \cdot \rho(a,b) \;\forall\, a,b \in \mathbb{R}^2$

$$\Rightarrow B_\rho\left(x, \frac{\sqrt{2}}{2}\varepsilon\right) \subseteq B_d(x,\varepsilon) \Rightarrow B_\rho\left(x, \frac{\sqrt{2}}{2}\varepsilon\right) \subseteq U.$$

7. Let $X = \{1,2,3\}$. What are some of the topologies on X.

$T_1 = \{\{1\}, \phi, X\}$, $T_3 = \{\{1\}, \{2\}, \{1,2\}, \phi, X\}$

$T_2 = \{\{1,2\}, \phi, X\}$ $T_4 = \{\{1\}, \{2,3\}, \phi, X\}$

$T_5 = \{\{1,2\}, \{2,3\}, \{2\}, \phi, X\}$.

$T_6 = \{\{1\}, \{1,2\}, \phi, X\}$.

$T = \{\{1,2\}, \{2,3\}, \phi, X\}$?

Remark: There are many topologies on a set. If $\mathcal{T}_1$ & $\mathcal{T}_2$ are topologies on a set X & $\mathcal{T}_1 \subseteq \mathcal{T}_2$, then $\mathcal{T}_2$ is finer (stronger, larger) than $\mathcal{T}_1$ & $\mathcal{T}_1$ is weaker (coarser) than $\mathcal{T}_2$.

8. Let X be a set and let $\mathcal{T} = \{U \in \mathcal{P}(X) : U = \phi$ or $X - U$ infinite$\}$.

It is called finite complement topology.

If X is finite then the finite complement topology on X is a discrete topology.

Finite Complement topology on $\mathbb{R}$.

$$\mathcal{T} = \{U \subseteq \mathbb{R} : \mathbb{R} \setminus U \text{ is finite, or } U = \phi\}$$

Open sets: , $\mathbb{R}$, $\mathbb{R}\setminus\{1\}$, $\mathbb{R}\setminus\{1,2,3\}$, All sets missing only finitely many points

For $a \in \mathbb{R}$, the smallest open nbd of a is $\mathbb{R}\setminus F$ for some finite set $F \subseteq \mathbb{R}\setminus\{a\}$.

9. Let X be a set and let $\mathcal{T} = \{ U \in \mathcal{P}(X) : U = \phi$ or $X - U$ is Countable$\}$

It is called the countable complement topology on X.

Basis for a Topology

A topology on a set can be a complicated collection of subsets of a set X, and it can be difficult to describe the entire collection. In most cases one describes a subcollection that "generates" the topology.

One such subcollection is called a Basis.

Defn: Let $(X, \mathcal{T})$ be a topological space. A basis for $\mathcal{T}$ is a subcollection $\mathcal{B}$ of $\mathcal{T}$ with property that if $U \in \mathcal{T}$ then $U = \phi$ or there is a subcollection $\mathcal{B}'$ of $\mathcal{B}$ s.t $U = \bigcup\{B : B \in \mathcal{B}'\}$.

Examples:

1. $\mathcal{B}$ = collection of all open intervals is a Basis for the usual top. on $\mathbb{R}$.

2. If X is a set, then $\mathcal{B} = \{\{x\} : x \in X\}$ is a basis for the discrete topo. on X.

3. Let (X,d) be a metric space. Then the family $\mathcal{B} = \{B(x;\varepsilon) : x \in X, \varepsilon > 0\}$ is a basis for the topo. generated by d.

4. Let $X = \{1,2,3\}$, and $\mathcal{B} = \{\{1,2\}, \{2,3\}, X\}$

Suppose $\mathcal{B}$ is a Basis for a Top. on X. Then by def. $\mathcal{B} \subseteq \mathcal{T}$. Hence $\{1,2\}, \{2,3\} \in \mathcal{T} \Rightarrow \{1,2\} \cap \{2,3\} = \{2\} \in \mathcal{T}$.

But $\{2\} \neq \phi$ and no subcollection $\mathcal{B}'$ of $\mathcal{B}$ s.t. $\{2\} = \cup\{B : B \in \mathcal{B}'\}$. Hence $\mathcal{B}$ is not a Basis for $\mathcal{T}$.

Closed Sets, Closures, and Interiors of sets

<u>Defn</u>: A subset A of a topological space $(X,\mathcal{T})$ is closed provided its complement, $X-A$ is open.

<u>Example</u>

1. A closed interval $[a,b]$, $a<b$ in $\mathbb{R}$ is closed with respect to the usual topology on $\mathbb{R}$, since $\mathbb{R} \setminus [a,b]$ is $\underline{(-\infty, a)} \cup \underline{(b,\infty)}$.

If $a \in \mathbb{R}$, $\{a\}$ is closed with respect to the usual topo. on $\mathbb{R}$, because $\mathbb{R} \setminus \{a\} = (-\infty, a) \cup (a,\infty)$

- With respect to the finite complement topology on X, a proper subset of X is closed iff it is a finite set

2. If (X, d) is a metric space, $x \in X$, and $\varepsilon > 0$, then $B(x, \varepsilon)$ is an open set.

$A = \{y \in X : d(x,y) \leq \varepsilon\}$ is a closed subset of X.

Pf: To show A is closed, we show that $X \setminus A$ is open.

Let $y \in X \setminus A$ & let $r_y = d(x,y)$. Then $r_y > \varepsilon$

$\Rightarrow$ $\delta_y = r_y - \varepsilon$ is a positive number. Thus to claim $B(y, \delta_y)$ is an open set $\subseteq X - A$. Suppose $z \in B(y, \delta_y)$.

Then $d(y,z) < \delta_y$. Now $d(x,z) + d(z,y) \geq d(x,y)$

$\Rightarrow$ $d(x,z) \geq d(x,y) - d(z,y) > d(x,y) - \delta_y = r_y - \delta_y = \varepsilon$.

$\Rightarrow$ $d(x,z) > \varepsilon$ & hence $z \notin A$. Therefore, $B(y, \delta_y) \subseteq X - A$.

Theorem: Let $(X, \mathcal{T})$ be a topological space. Then the following conditions hold:

(a) X and ϕ are closed sets

(b) If $\mathcal{A}$ is a finite family of closed subsets of X, then $\cup\{A : A \in \mathcal{A}\}$ is closed set.

(c) If $\mathcal{A}$ is a family of closed subsets of X, then $\cap\{A : A \in \mathcal{A}\}$ is a closed set.

Defn: The closure $\overline{A}$ of a subset A of a topological space $(X,\mathcal{T})$ is the intersection of the closed sets that contain A.

Note: $\overline{A}$ is ~~the~~ "smallest" closed set that contains A, i.e; B is any closed set s.t $A \subseteq B$, then $\overline{A} \subseteq B$

Theorem: Let A be a subset of a topological space $(X,\mathcal{T})$, and let $x \in X$. Then $x \in \overline{A}$ iff every nbd of x has a non-empty intersection with A.

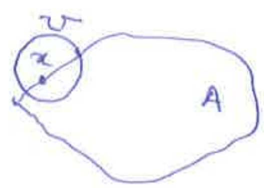
Pf: Suppose $x \notin \overline{A}$. Then there is a closed set C such that $A \subseteq C$ & $x \notin C$. $\Rightarrow$ $X - C$ is a nbd of x such that $(X-C) \cap A = \phi$

Suppose there is a nbd U of x such that $U \cap A = \phi$. Then $X - \cancel{A}\,U$ is a ~~b~~ closed set s.t $A \subseteq X-U$ & $x \notin X - U$. Hence $x \notin \overline{A}$.

Remark: If (X,d) is a metric space & $A \subseteq X$, then it follows from the above theorem that $x \in \cancel{B}\ \overline{A}$ iff for each positive $\varepsilon > 0$, $B(x,\varepsilon) \cap A \neq \phi$.

Defn: Let A be a subset of a topological space $(X,\mathcal{T})$. A point $x \in X$ is a limit point A, provided every ~~point~~ nbd of x contains a point of A that is different from x. Let A' denote the set of all limit points of A.

$U \cap (A \setminus \{x\}) \neq \phi$



Examples:

1. With respect to the usual topology on $\mathbb{R}$, if $a, b \in \mathbb{R}$ & $a<b$, then a is a limit point of each of $[a,b]$, $[a,b)$ & (a,b).

Thm: Let A be a subset of a topological space $(X, \mathcal{T})$. Then $\bar{A} = A \cup A'$.

Pf: Let $x \in \bar{A}$. If U is a nbd of x, then $U \cap A \neq \phi$. If $U \cap A = \{x\}$, $\Rightarrow x \in A$. If every nbd of x contain a point of A different from x, then $x \in A'$. In either case, $x \in A \cup A'$, so $\bar{A} \subseteq A \cup A'$.

Suppose $x \in A \cup A'$. If $x \in A \Rightarrow x \in \bar{A}$. If $x \in A'$, then every nbd of x contains a point of A that is different from x. $\Rightarrow x \in \bar{A}$. Thus $A \cup A' \subseteq \bar{A}$

Thm: Let A be a subset of a topo. space $(X, \mathcal{T})$. Then A is closed iff $A' \subseteq A$.

Boundary of a Set:

Defn: Let A be a subset of a topological space $(X, \mathcal{T})$. A point $x \in X$ is a boundary point of A provided $x \in \bar{A} \cap \overline{X-A}$. The boundary of A $\boxed{bd(A)}$, is the set of all boundary points of A.

Defn: A subset A of a topological space $(X, \mathcal{T})$ is dense in X provided $\overline{A} = X$.

If X has a countable dense subset, then $(X, \mathcal{T})$ is a seperable space.

Example:

1. Let $(\mathbb{R}, \mathcal{T})$ be the usual topology on $\mathbb{R}$. Then

$\uparrow$ topo.

the set of rational numbers is a countable dense subset & hence $(\mathbb{R}, \mathcal{T})$ is a separable space.

2. Let $\mathcal{T}$ be the finite complement topology on $\mathbb{R}$.

Then every infinite subset is dense in $\mathbb{R}$.

Let A be an infinite subset of $\mathbb{R}$. Let $x \in \mathbb{R}$ & let U be a nbd of x. Since $\mathbb{R} - U$ is finite, U must contain all but a finite numbers of members of A.

In particular, $A \cap U \neq \phi$. Therefore $x \in \overline{A}$ & hence $\overline{A} = \mathbb{R}$.

Defn: The interior, $\text{int}(A)$, of a subset A of a topological space $(X, \mathcal{T})$ is the union of the open sets that are subsets of A.

Defn: Let A be a subset of a topo. space $(X, \mathcal{T})$ & let $x \in A$. Then x is an interior point if there is a nbd U of x s.t $U \subseteq A$.

§ Convergence

Recall

$f: \mathbb{R} \to \mathbb{R}$ is continuous at a point "x" in $\mathbb{R}$ iff whenever $\{x_n\}$ is a sequence in $\mathbb{R}$ that converges to x then $\{f(x_n)\}_n$ converges to $f(x)$.

Defn: A sequence $\{x_n\}$ of points in a topological space (X, Γ) converges to a point x in X provided that for each nbd U of x there is a natural number N s.t $x_i \in U$ $\forall$ $i \geq N$.

Example:

1. Let Γ be the trivial topology on a set X, let $\{x_n\}$ be a sequence in X & let x be any member of X. Then $x_n \to x$.

   Indeed, let $\{x_n\}$ be a seq in X, let $x \in X$ & let U be a nbd of x. Since Γ is a trivial topo. & $x \in U$, $U = X$.

   Therefore $x_n \in U$ for every $n \in \mathbb{N}$.

2. Let Γ be the countable complement topology on $\mathbb{R}$, and let A be the set of irrational numbers. Then $0 \in \overline{A}$, but there is no seq in A that converges to 0.

Pf: Let U be a nbd of 0. Then $\mathbb{R} - U$ is countable. Thus, since A is uncountable, $A \subsetneq \mathbb{R} - U$. Therefore $U \cap A \neq \phi$, & so $0 \in \overline{A}$. Let $\{x_n\}$ be any seq in A. Then $\mathbb{R} - \{x_n : n \in \mathbb{N}\}$ is a nbd of 0 that does-not contain any point $\{x_n\}$. Therefore $\{x_n\}_n$ does-not converge to 0.

Defn: A seq. $\{x_n\}_n$ in a metric space (X, d) is a Cauchy seq. provided for each $\varepsilon > 0$, $\exists$ N s.t if $m, n \geq N$, then $d(x_m, x_n) < \varepsilon$.

Theorem: Let (X, d) be a metric space & let $\{x_n\}_n$ be convergent in X. Then $\{x_n\}_n$ is a Cauchy sequence.

pf: Let $x_n \to x$, Let $\varepsilon > 0$, since $x_n \to x$, $\exists N$ s.t $x_i \in B\left(x, \frac{\varepsilon}{2}\right)$ $\forall$ $i \geq N$.

Suppose $m, n \geq N$. Then $d(x_m, x) < \frac{\varepsilon}{2}$ & $d(x_n, x) < \frac{\varepsilon}{2}$.

Therefore, $d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$.

Qn: Every Cauchy seq. is convergent in (X, d)?

Example: Let d be the usual metric on $\mathbb{R}$, let $X = \mathbb{R} - \{0\}$, & let $\rho = d|_X$. Let $x_n = \frac{1}{2^n}$ Then $\{x_n\}$ is a Cauchy in (X, ρ), yet it does-not converge.

Defn: A metric space (X, d) is complete provided that every Cauchy sequence in X converges.

<u>Theorem:</u> Let (X,d) be a metric space such that every Cauchy seq in X has a convergent subsequence. Then (X,d) is complete.

<u>pf:</u> Let $\{x_n\}_n$ be a Cauchy sequence, and let $\{x_{n_i}\}$ be a subsequence of $\{x_n\}_n$ that converges to x.

<u>Claim:</u> $x_n \to x$. Let $\varepsilon > 0$, $\exists N$ s.t if $m, n \geq N$, then $d(x_m, x_n) < \frac{\varepsilon}{2}$. Since $x_{n_i} \to x$, there is a natural number i_p s.t $i_p > N$ & $d(x_{i_p}, x) < \frac{\varepsilon}{2}$.

Then $d(x_n, x) \leq d(x_n, x_{i_p}) + d(x_{i_p}, x) < \varepsilon \quad \forall n \geq N$.

- If d is the usual metric on $\mathbb{R}$, then $(\mathbb{R}, d)$ is complete.

§ Continuous function:

<u>Recall:</u>

<u>Thm:</u> Let $(X, \mathcal{T})$ & $(Y, \mathcal{U})$ be topological spaces & let $f : X \to Y$. Then the following statements are equivalent:

(a) f is continuous

(b) If $V \in \mathcal{U}$, then $f^{-1}(V) \in \mathcal{T}$.

(c) If C is a closed subset of $(Y, \mathcal{U})$, then $f^{-1}(C)$ is closed subset of $(X, \mathcal{T})$.

<u>pf:</u> (b) $\Rightarrow$ (c). Let C be a closed subset of $(Y, \mathcal{U})$. Then $Y - C \in \mathcal{U}$, by (b) $f^{-1}(Y - C) \in \mathcal{T}$.

now, $f^{-1}(C) = X - f^{-1}(Y - C)$. Therefore $f^{-1}(C)$ is a closed subset of $(X, \mathcal{T})$.